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PI Name:	Badler, Norman I Ph.D.		
Project Title:	Crew Task Simulation for Maintenance, Training, and Safety		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
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Human Research Program Elements:	None		
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Zip Code:	19104-6389	**Congressional District:**	2
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Contact Email:			
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	Ambitious space missions will present new challenges for space human factors as crews interact with and maintain complex equipment and automation. This demands that procedures for task design, rehearsal, and procedure refreshing be based on on-ground validated procedure instructions. We investigated requirements for formulating, interpreting, and validating procedure instructions. Crew instructions should express clearly and unambiguously complex actions and their expected results. As NASA moves toward a “what to do” rather than a “how-to-do” approach, it is essential that textual instructions convey what is meant and correlate with equipment function and construction. Natural language instructions must coordinate with graphical simulations to provide both textual and visual guidance and context-sensitivity for procedures, especially when the equipment functions and features are accessible to the simulation. Current manuals and procedures need all alternatives to be made explicit, leading to clumsy documentation which must be laboriously prepared in advance. Any documentation for a complex procedure should “understand” the equipment and the human factors of the crew. One important aspect of “what-to-do” procedure documentation is that purposes and		

Task Description:

termination conditions of continuous processes are very frequently cited. These need to be generated from or interpreted by simulated actions. Although our focus in this project has been on ground-based procedure execution and validation, we envisioned future applications to spaceflight crews who will use natural language to communicate with each other, the on-board automation, and to training and refresher simulations. Our specific research goals were to use various user interfaces -- such as natural language instructions or virtual reality -- to control virtual crewmembers for procedure formulation, validation, inflight training, crew task allocation, and unusual procedure simulation. This project had its genesis in the confluence of our prior research efforts for NASA, the Air Force, and the National Science Foundation (NSF). Having investigated computer graphics digital human models for quite some time, and having been familiar with their use for ergonomics and human factors applications, we understood some of the compelling problems that today still constrains effective and widespread utilization of those models.

• From prior NASA research we had an understanding of spaceflight applications for human models, and the difficulties in modeling in-flight biomechanics, micro-gravity actions, and complex maintenance task manual manipulations. We began this project with the belief and expectation that digital human models could simulate spaceflight procedures to insure human competency to perform similar tasks in reality. Such simulations would insure good pre-flight designs, but required significant software investment and proper utilization by craft contractors. Indeed, some ergonomics human modeling tools do exist and are used in the design stages of a major project, but most of these are interactive systems and are not structured for flexible simulation: they are not particularly capable of taking instructions and executing them in a variety of design alternatives, and considerable operator expertise is required to set-up and run an analysis.

• From prior Air Force research, we did extensive studies of instruction manuals for both aircraft maintenance and other mechanical applications. From this work we gained an understanding of the existing state-of-the-art in automated (electronic) instructions, and a priority ordering of problems that need to be addressed in order to produce better electronic instruction validation. The roles of validation and simulation are intertwined: a simulation may or may not succeed, and if not, the causes of failure must be established and understood. Failures may stem from or involve accommodation problems due to anthropometric variations, equipment design flaws, unforeseen procedure steps, impossible equipment access, misunderstanding or ignorance of equipment function, or inadequate or ambiguous instructions. Validation requires an integration of human modeling with equipment functional simulation, and a more automated analysis framework in which to probe the space of success and failure modes.

• From prior NSF research we studied the interrelationships between natural language and human animation, and developed a Parameterized Action Representation (PAR) and an Actionary of digital definitions suitable for an action database. This work focused on how manual procedure instructions might be set on a computational basis appropriate to both natural language processing and computer animation on virtual human models. PAR describes actions with parameters such as agent, object, motion qualities, paths, preparatory specifications, termination conditions, and sub-actions so they can be re-used in a wide variety of spatial contexts, by various virtual actors, and with action modifiers.

This NASA project sought to combine these observations, knowledge, and methodologies to serve the objective of designing better instruction delivery, simulation, and validation tools. At the conclusion of the project, we have three completed PhD dissertations to show as specific accomplishments, and much other work, both reported and ongoing, that directly derived from NASA support. In the following sections, we sketch some of these results and their significance. This project has led to several specific accomplishments. We summarize each in this section.

Automated constrained human arm reach

We have shown, in Ying Liu's PhD thesis, a novel way of computing real-time human arm reach actions for maintenance, operation, or scientific tasks in constrained physical environments. Such movements are essential for computer software that must automatically synthesize complex arm motions in confined spaces, e.g., in a glovebox. Rather than rely on a human operator to guide the synthetic reach, the program can do the analysis itself, and do so in a fashion that:

1. Uses a simple spatial partition and spatial search algorithm to move the hand through space toward the goal,

2. Avoids arm and body contacts with the environment using the graphics hardware board to do fast collision detection,

3. Does not use the traditional configuration-space planning approach common in robotics, and

4. Uses arm strength data to favor more natural (less awkward) shoulder, elbow, and wrist postures during solution search.

This work is being extended to incorporate better empirical data (reported by others) to extend the reach chain beyond the shoulder joint to the whole upper body.

Shape Preserving object simplification

In Koji Ashida's PhD thesis, we began an investigation of the very general question of converting geometric models from complex mega-polygon meshes into more compact representations that automatically preserved shape features such as pointed parts or sharp edges. The motivation for this arises from placing a simulated human (maintainer) in a complex modeling 3D environment. The numerous surfaces needed in a polygon model mostly arise from visual needs and local details that are possibly not very relevant to the accomplishment of a maintenance task. But some features, such as points and sharp edges, are important for safety. Semantically careless polygon decimation can eliminate some of these features and affect task accomplishment methods or viability. Although we have not been able to apply these results directly, the issues are nonetheless important for both equipment design and automatic human task simulation. In the former, the design features at the human manipulation level of detail are important, but make be overwhelmed by essential engineering detail. In the latter, fewer polygons means faster computation of reach and positioning tasks, since collision detection costs are highly dependent on the mesh size of the 3D model.

Interactive vibrotactile experience for confined arm reach

A non-automated approach to task validation motivated Aaron Bloomfield's PhD. For maintenance and operational training, a virtual environment can provide visual feedback, but a human operator will not have the tactile benefit of feeling the mechanical constraints in, say, a confined space such as an equipment bay or a glove box. Here we explored the question of whether a physical stimulus in a virtual reality experience could effectively substitute for a physical obstacle. This is the interactive counterpart to Ying Liu's work: the reach computation is performed by a real person but

in a virtual environment. The physical component that is missing is the collision detection and response, so we use an array of vibrotactile devices (tactors) on the arm to simulate the “feel” of touching a real object though none is physically present. These small, inexpensive computer-controlled tactors deliver vibration to a small area of skin. We position up to 24 of these on the arm and hand of a subject. As the subject navigates a confined virtual space she feels the vibratory sensations wherever her arm or hand penetrates the virtual models. In experiments, users were able to perform reaches to occluded targets in confined volumes faster with the vibrotactile display on their arm. Such a haptic array may help motor learning of complex tasks and provide valuable physical feedback (though not physical constraint) during initial task learning or later rehearsal. This could translate to reduced training time for complex assembly or disassembly tasks, with the added benefit that the human subject must actually perform the task without having the cost or expense of a physical model.

Deformable arm model with biomechanical tissue properties

In Hogeun Shin’s work, we created a human arm model with biomechanical deformation parameters. Using a novel device to measure a local pressure/displacement function, the indentation of the arm tissue can be simulated when an external blunt (spherical) obstacle is applied. The purpose of such a model is to see whether the deformation of soft tissues can affect the reachable space of the arm, since all existing ergonomic digital human models have rigid surfaces. Reach analyses performed with rigid body collision detection are apt to be conservative (underestimating reachable space) with respect to what a real (deformable) arm model can reach. In informal experiments it was found that reach could be extended up to about 15% if deformation over small obstacles was considered. Results for larger obstacles (such as flat areas) will be closer to the predictions from rigid geometry models since volume-preserving tissue displacements limit the travel of the obstacle into the arm space. Accordingly, this research showed a local effect but not enough to continue with more detailed deformable arm models at this time. It is possible than these ideas can be profitable applied to lower torso for whole body maintenance access problems or buttocks deformations for optimizing seat design, but to date we have not specifically addressed data acquisition methods for these body regions nor these particular applications.

The personality factor

As we investigated the possibilities for crew simulation, it became clear that very important factors to consider in task accomplishment and success were an individual’s personality and psychological state. This might even be the most important subjective issue for long duration flights: actually having the crew do what they need to do and not do what they should not do. Understanding how to include these considerations in task, workload, and mission simulations would presumably lead to better working conditions and overall mission success. Jan Allbeck’s approach to this difficult aspect of human behavioral modeling was to look at a classification of plotlines from narrative stories. These plotlines could provide a “backbone” of human actions and reactions to support the selection of individual tasks. Plot action choices are highly dependent on an individual’s personality, culture, and motivators. Work plans could then seek to include countermeasures to mitigate plotlines that might lead to negative crew consequences.

Our preliminary explorations led to interaction with a Penn colleague, Professor Barry Silverman. Dr. Silverman’s work has produced a human performance model (PMFserv) that includes physiological factors (hunger, thirst, and fatigue), psychological factors (personality and emotions), and cultural factors (values and standards) and personal motivators and goals. We also examined work by David Musson at UT Austin, on astronaut and long-duration crew interpersonal relationships. The result of this investigation was a joint proposal by Musson, Silverman, and Badler to the NSF (still pending) to incorporate personality testing instrument results into PMFserv and use these to simulate possible crew interactions.

Parameterized actions and the Actionary

We have continued to develop our parameterized action representation (PAR) and simulation system that is based on it. Our aim was to create a system in which domain experts, not simulation experts, can build a simulation to test maintenance procedures, habitability, or work schedules. Our previous versions of the PAR system were based on Python and C++ under the IRIX operating system. In order to optimize the code and make it more accessible to a larger community, we have been porting the code to a Windows based operating system and used a commonly-used free database called MySQL to store the database of available actions, the Actionary. PAR data is entered into MySQL through a graphical user interface (GUI) consisting of various windows. Through the main windows, the Object Edit and Action Edit windows, the user can create new objects, agents, and actions, as well as edit, and search through those already defined in the database. The programming of flexible base dialogs for repeated functionality among objects and actions (e.g., coordinate entry, object/action search dialogs) allows the user to quickly become accustomed to the interface, while the size of the program stays minimal. The object, agent, and action data are stored in a relational MySQL database in a normalized structure such as to speed up the retrieval of objects, agents, and actions. Also, because of the possibility of integrating this data into a mobile environment, the database model has been tuned to efficiently occupy a minimal storage space. This work has influenced future instruction delivery models for NASA, since the PAR allows re-use of stored instructional components and detail level-dependent delivery of instructional materials.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our research seeks to improve the analysis of human task performance in space-related activities. In order to perform task analyses automatically, we have developed a fast human arm constrained reach algorithm. This arm control algorithm would also ease the analyst’s role in determining human strategies for achieving constrained reaches. By speeding up the analysis process, the human factor engineer can examine more situations, improving efficiency and expanding the coverage of analyzed tasks. We have also investigated this problem from the virtual reality simulation side, and have shown significant training opportunities for vibrotactile devices to aid human performance in complex manual manipulation tasks. Such ergonomic analysis benefits are not confined to aerospace applications.
By beginning to look into psychological and personality factors in task performance we are addressing what might be the most important subjective issue for long duration flights: actually having crew do what they need to do and not do what they should not do. This would support more successful missions and could be applicable to earth as well as space applications.

Finally, the generation of performable and validatable instructions has led to a subsequent NASA-funded project to

transition some of these experiences and knowledge into the next generation instruction delivery system for future NASA spaceflight missions.

Task Progress:

This project has led to several specific accomplishments. We summarize each in this section.
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